

Thesis Progress Report – Year²

Date: 10/06/2026

Name and first name of the doctoral candidate: MA JUN

Research unit: LMAC (Laboratoire de Mathématiques Appliquées de Compiègne)

Thesis supervisor(s): Faker Ben Belgacem & Faten Jelassi

This report helps you to present your thesis at the annual meeting of your Individual Monitoring Committee (CSI), both HR (Human Resources) and Scientific. You must forward this report to your CSI members at least 10 days prior to your CSI meeting date.

You must also upload this report in ADUM in your administrative re-registration file.

Here are at least the topics to fill in, you can add some if you wish:

Brief description of thesis topic and objectives:

This project aims to develop innovative hybrid thermal energy storage (TES) systems by integrating topology optimization, shape optimization, and reduced-order modeling, complemented by the potential resources and applications of physics-informed deep learning techniques. More specifically, the objective is to enhance the thermal conductivity of TES systems incorporating phase change materials (PCMs), by overcoming current limitations through advanced design strategies and numerical simulation. Particular emphasis will be placed on the encapsulation of TES materials using thermally conductive agents, in order to optimize heat transfer and thereby accelerate the charging process and, more importantly, the extraction of heat from the storage medium during short charging/storage/discharging cycles.

It is generally acknowledged that the optimization problems associated with charging and discharging should not lead to similar, or even nearly identical, capsule distributions. In this context, this issue should be carefully investigated in order to design an intermediate composite medium that enables efficient heat input and easy thermal evacuation. One possible approach to making an informed and optimal decision is to incorporate multi-objective optimization tools, such as the Pareto principle. The implementation of topology optimization procedures will make it possible to explore reduced-order models for efficient capsule arrangements, preserving the high thermal capacity of phase change materials (PCMs) while improving the overall thermal conductivity. This will facilitate fast numerical simulation and analysis of these complex systems. PINN algorithms will therefore be investigated for their potential to predict system performance and optimize design configurations.

Summary of work done for the past year and background:

The general objective of the thesis is to study mathematical and numerical models for heat transfer problems arising in phase-change materials and thermal storage systems. In such systems, the presence of interfaces between different materials, inclusions, or phases may strongly influence the effective thermal behavior. In particular, imperfect thermal contact can lead to a temperature jump across the interface, which is commonly modeled through Kapitza-type interface conditions. These interface conditions provide a relevant mathematical framework for describing thermal resistance effects in composite or heterogeneous media.

During the past year, the work has developed along three complementary directions. The first direction concerns the numerical investigation of Kapitza-type interface problems. The second direction concerns topology optimization for heat transfer problems, based in particular on the study of

topological derivatives. The third direction concerns preliminary implementations of XPINN methods for variable-coefficient heat conduction equations, as a first step toward physics-informed approaches for heterogeneous thermal media.

For the Kapitza-type interface problems, the main effort has been devoted to the development and testing of computational tools allowing the simulation of both linear and nonlinear interface laws. In this context, I implemented numerical frameworks based on FreeFEM and Matlab in order to solve model interface problems, post-process the results, and analyze the behavior of the computed solutions under different parameter regimes.

A significant part of this work concerned nonlinear interface conditions. Numerical experiments showed that standard Newton-type methods and classical parameter-continuation strategies may become insufficient when the nonlinear interface law generates multiple solution branches or turning points. In particular, the computations revealed bifurcation-type behavior and sensitivity with respect to the initial guess and continuation parameter. These observations motivated the use of pseudo-arclength continuation, which provides a more robust numerical approach for following solution branches through folds and for capturing branches that are difficult to obtain by direct parameter continuation.

In parallel, I also studied the basic theory and numerical use of topological derivatives in the context of heat transfer and topology optimization. This part of the work aimed at understanding how the insertion of small inclusions or material perturbations affects a thermal objective functional. I carried out numerical verification and implementation of topological derivative-based criteria, with the objective of identifying favorable regions for material modification or inclusion insertion. These first numerical experiments provided useful insight into how sensitivity information can be used to guide topology optimization procedures for thermal systems.

A further part of the work concerned the use of physics-informed neural network methods, especially XPINN, for heat conduction problems with spatially varying thermal conductivity. I successfully implemented XPINN-based solvers for both steady variable-coefficient heat conduction equations and time-dependent variable-coefficient heat conduction equations. In these models, the thermal conductivity differs from one subdomain to another, which makes them a natural preliminary setting for heterogeneous media and interface-type problems. These implementations allowed me to test domain-decomposition-based neural network methods on heat transfer problems and to assess their potential for future applications to Kapitza-type interfaces and topology optimization.

The numerical work carried out during this period has therefore played several roles. First, it provided practical computational tools for simulating Kapitza-type interface problems and visualizing their solution structure. Second, it helped identify the main mathematical and numerical difficulties associated with nonlinear interface laws, especially possible non-uniqueness, branch selection issues, and convergence difficulties of standard iterative solvers. Third, the work on topological derivatives introduced a complementary optimization-oriented perspective, which is directly connected to the design of efficient thermal storage systems and optimized material distributions. Finally, the XPINN implementations provided a preliminary physics-informed computational framework for heterogeneous heat conduction problems and opened the possibility of combining neural-network-based solvers with interface modeling and topology optimization.

In parallel, I have started consolidating the theoretical background related to the linear Kapitza problem, including the weak formulation, the role of interface terms, and the corresponding finite element discretization. This linear framework is an essential foundation for the continuation of the thesis, since it provides the reference setting from which nonlinear interface models can be studied more systematically.

Overall, the work of the past year has allowed the thesis direction to become more focused. On the one hand, the numerical study of Kapitza-type interface problems has highlighted the importance of nonlinear interface laws, bifurcation phenomena, and continuation methods. On the other hand, the study and implementation of topological derivative-based methods has opened a second research direction related to topology optimization for heat transfer and thermal storage applications. In addition, the successful implementation of XPINN methods for steady and time-dependent variable-coefficient

heat equations provides a methodological basis for future physics-informed approaches to heterogeneous media, interface problems, and optimization-driven simulations. The next stage of the thesis will aim to strengthen the connection between these numerical developments and the underlying mathematical analysis, while progressively integrating finite element methods, continuation techniques, topological sensitivity analysis, and PINN/XPINN-based tools.

Perspectives and projects on thesis work for the coming year:

The work planned for the coming year will be organized around two complementary research directions.

The first concerns the mathematical and numerical analysis of Kapitza-type interface problems, while the second focuses on topology optimization for thermal storage systems, with the possible use of PINN and XPINN methods as complementary computational tools. These two directions remain connected by the general objective of the thesis: the development of mathematical and numerical models for heat transfer problems in heterogeneous media and phase-change material systems.

The first objective will be to consolidate the linear finite element framework for Kapitza-type problems. This includes a more systematic study of the weak formulation, the role of the interface discontinuity condition, the structure of the finite element discretization, and the connection between the continuous formulation and the implemented numerical schemes. This step is necessary in order to establish a reliable reference model before addressing nonlinear extensions. Particular attention will be paid to the consistency between the mathematical formulation and the FreeFEM implementation.

The second objective will be to continue the study of nonlinear Kapitza-type interface laws. The numerical experiments carried out so far indicate that nonlinear interface conditions may generate multiple solution branches, turning points, and convergence difficulties for standard Newton-type methods. In the coming year, I plan to analyze these phenomena more systematically by combining nonlinear finite element discretizations with continuation strategies. In particular, pseudo-arclength continuation will remain an important tool for following solution branches through folds and for studying bifurcation-type behavior.

In parallel, a second important research direction will concern topology optimization for heat transfer and thermal storage applications. This part will rely on the analysis of topological derivatives and shape derivatives, with the aim of designing optimized material distributions or interface configurations that improve the thermal performance of the system. The numerical implementation of these optimization methods will be further developed and integrated into the finite element simulations already carried out in FreeFEM and Matlab.

PINN and XPINN methods will also be studied in this context. Rather than replacing the finite element framework, these methods will be considered as complementary tools for solving, approximating, or accelerating certain heat transfer and optimization problems. In particular, XPINN-type domain decomposition strategies may be relevant for interface problems, while PINN-based approaches may provide flexible surrogate models for the repeated simulations encountered in topology optimization. This direction will be developed in connection with the work on topological and shape derivatives, and under the guidance of the thesis co-supervision.

Another objective will be to clarify the respective roles of classical finite element methods and physics-informed neural network approaches. The finite element framework provides a robust and mathematically interpretable reference method, while PINN/XPINN approaches may offer greater flexibility for complex geometries, interface conditions, or parametric studies. A thorough comparison of these approaches will make it possible to determine which parts of the thesis should be developed as rigorous finite element analysis and which parts can be treated as numerical or data-driven extensions.

The objective for the coming year is to obtain a clearer and more coherent thesis structure. Ideally, this work should lead to a first article or an in-depth chapter on the linear finite element formulation of the Kapitza problem and its numerical implementation, as well as to a second research direction devoted to

nonlinear interface laws and bifurcation phenomena. In addition, topology optimization, based on topological and shape derivatives and potentially supported by PINN/XPINN methods, should be developed into a structured numerical framework for thermal storage applications. These elements will form the basis for the final stages of the thesis, which will be devoted to completing the nonlinear analysis, preparing publications, and writing the doctoral dissertation.

Doctoral training:

- [1] CP36 - Practical tools to give meaning to your work and solve problems (6h)
- [2] CST02 : Méthodes pour la modélisation aléatoire (20h)
- [3] LA91, niveau A1/A2, - Semestre de printemps 2025 (15h)
- [4] CST02 Python scientifique et Jupyterlab (26h)
- [5] CIMPA research school: Mathematics, Artificial Intelligence and Application 2025(48h)
- [6] CST02: Advanced Mechanics of Solids, Fluids, Structures and Flexible Multibody Systems(35h)

Review of dissemination actions and scientific and technical productions: (specify: submitted or accepted)

* **Internal scientific presentations** (specify dates and audience), oral or poster

NON

* **Publications in journals with or without a reading committee** (specify title, authors, references)

ACL: Articles in international or national peer-reviewed journals listed in international databases (ISI Web of Knowledge, Scopus, Pub Med...).

ALC: Articles in refereed journals not listed in international databases.

ASCL: Articles in non-peer-reviewed journals.

[ACL] Bejaoui, E., Belgacem, F. B., Jelassi, F., & Ma, J. (2026). Steady conduction problems with non-linear Kapitza contact resistance: Existence and non-uniqueness. *Mathematics and Computers in Simulation*.

[ACL] Bejaoui, E., Ben Belgacem, F., Jelassi, F., & Ma, J. (2026). Semilinear elliptic problems with non-linear contact resistance in composite domains. HAL preprint, hal-05659943. <https://utc.hal.science/hal-05659943>

* **Communications to symposia/congresses with acts** (specify title, dates, place, title, authors and references)

ACTI: Communications with acts at an international congress.

ACTN: Communications with deeds at a national congress.

COM: Oral communication without acts at an international or national congress.

POST: Posted communications at an international or national conference.

* **Patent filings** **NON**

* **Awards and honors** **NON**

* **Participation in the organization of national/international symposia** **NON**

* **Involvement in the life of the institution, in bodies ...** **NON**

* **Public communication actions** (specify the name, date and place of the meeting and your type of action) (e.g. Science Day) **NON**